S2 Supplementary Data

Canonical source freeze for every empirical number in this article

Article type: Supplement

Short title: S2 Data - NDT

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Source-binding rule

Every empirical number in the main text must point to one canonical companion artifact and, where relevant, one frozen result bundle. Intermediate or superseded merges are not used. This table is the freeze.

Anchor	Canonical article	Frozen bundle / contract	Numbers used here
Spindles	Sleep Spindles Briefly Shift Cortical Network Geometry in Two Independent Overnight EEG Datasets	ANPHY $N = 27$; BOAS/ ds005555 $N = 128$; Rich-Sleep $N = 24$; Pass A/B/C 2026-07-03	Timing d_n: ANPHY $+0.106$ ($q = 0.009$), BOAS $+0.165$ ($q = 0.011$); matched-control specificity incomplete
Coma	Recovery-Linked Neural Geometry After Cardiac-Arrest Coma	I-CARE PhysioNet 2.1; `20260606_combined_invalidity_fix`	301 analyzable patients, 5322 runs; 281 CPC 1 vs 5 (141/140); persistence $d = -0.909$ and 0.896; tube $d = 0.838$ (0.786 in 0–1 h); AUROC 0.599 → 0.662
Meso–macro	Trial-aligned associations between Neuropixels proxies and surface EEG during cortical stimulation in head-fixed mice	DANDI 000458 Claar/ Rembado Neuropixels + surface EEG; NMD EEG-only local-linear chart; alignment incompletely validated	Median subject $r = 0.406$ (LFP SD with EEG SD) and $r = 0.395$ (spike rate with EEG spectral entropy), claim class 1, conditional on nominal event labels; residual-variance fraction 0.856 (archived diagnostic 0.8557), claim class 4 fail

Table 1: Canonical sources. Do not substitute a later or earlier bundle without rewriting both this table and the main-text sentences that cite it. Companion analyses: (Langell, 2026a); (Langell, 2026b); (Langell, 2026c). Dataset records remain (Wei et al., 2024), (López-Larraz et al., 2025), (Amorim et al., 2023), and (Claar et al., 2023).

What is not the coma source

A corrected-ComBat merge dated `20260602` reported 298 analyzable patients (278 in the extreme-outcome contrast in some ledgers), pooled `tube_d_eff_median` $d = 0.638$, and a predictive increment near AUROC 0.595 → 0.687. That merge is a documented intermediate. The canonical article in the canonical coma companion is the **invalidity-fixed** pooled synthesis of `20260606`, which restored 301/281 and the AUROC pair 0.599 → 0.662. Numbers in this article follow that companion, not the ComBat intermediate.

Coma layer summary (canonical article)

Copied from the frozen companion, not re-estimated here.

Layer	Pooled d	Role
persistence hysteresis log-det	-0.909	strongest reachability volume-dynamics separator
persistence log-det	$+0.896$	reachability volume companion
tube effective dimensionality	$+0.838$	lead dimensionality; $+0.786$ already in 0–1 h
MNJ 9D spectral radius	-0.603	aligned, cohort-heterogeneous
MNJ 3D Frobenius	-0.499	aligned, cohort-heterogeneous
MNPS 9D spread	-0.302	descriptive, not the lead

Table 2: Pooled CPC 1 versus CPC 5 anchors from the canonical coma article. Positive d means higher in CPC 1.

Spindle timing versus specificity

This article uses the **timing** estimand (event versus far pre-event baseline) as the primary spindle constraint. The event-versus-matched-N2 contrast is reported in the companion as incomplete: several endpoints are baseline-dominated, and d_n changes sign across datasets for that reason. Reachability was not computed. EAP moderation of the geometry effect is rejected under the primary protocol. Peak-phase respiratory locking is a separate RichSleep companion and is not a second theoretical anchor.

Dissociation matrix sources

Rows in the main-text dissociation table and Figure 5 are companion constraints, not additional anchors. Canonical sources are the corresponding companion articles. Anesthesia and epilepsy rows in Figure 5 remain unpaid (not tested), not hidden positives.

The three added matrix rows are likewise not fourth anchors.

Matrix row	Canonical companion	Numbers / ceiling
Expected-stimulus omission (VISp)	Allen Visual Behavior omission geometry (Langell, 2026d)	Median $\delta \approx 0.980$, 30/30 sessions; AVB-adapted chart; Jacobian, diffusion, and reachability untested
IBL chart readout	Discrete chart-transition readout (Langell, 2026e)	Population decoder PASS (median accuracy 0.588, $p_{\text{perm}} = 0.0030$); discrete-flow operator STOP (held-out residual ≈ 1.001)
Human MTL–scalp	DANDI 000574 Sternberg analogue (Langell, 2026f)	Primary pairs $r = 0.100$ and $r = 0.062$ fail conjunctive nulls ($n = 9$); design-specific negative, not a 000458 replication

Table 3: Matrix-only constraints imported from the m3-bridge freeze. They bound recoverability; they do not expand the three-anchor spine.

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